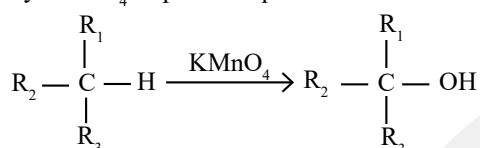


CHAPTER 10

Hydrocarbons

Alkanes

1. The alkane that can be oxidized to the corresponding alcohol by KMnO_4 as per the equation (2024 Re)



is, when:

- $\text{R}_1 = \text{H}; \text{R}_2 = \text{H}; \text{R}_3 = \text{H}$
 - $\text{R}_1 = \text{CH}_3; \text{R}_2 = \text{CH}_3; \text{R}_3 = \text{CH}_3$
 - $\text{R}_1 = \text{CH}_3; \text{R}_2 = \text{H}; \text{R}_3 = \text{H}$
 - $\text{R}_1 = \text{CH}_3; \text{R}_2 = \text{CH}_3; \text{R}_3 = \text{H}$
2. Given below are two statements: (2024)

Statement - I: The boiling point of three isomeric pentanes follows the order

$n\text{-pentane} > \text{isopentane} > \text{neopentane}$

Statement - II: When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- Statement - I is correct but Statement - II is incorrect.
 - Statement - I is incorrect but Statement - II is correct.
 - Both Statement - I and Statement - II are correct.
 - Both Statement - I and Statement - II are incorrect.
3. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is: (2023)
- 18
 - 16
 - 32
 - 30
4. The decreasing order of boiling points of the following alkanes is: (2022 Re)
- (a) heptane
 - (b) butane
 - (c) 2-methylbutane
 - (d) 2-methylpropane
 - (e) hexane

Choose the **correct answer** from the options given below :

- (a) $>$ (e) $>$ (c) $>$ (b) $>$ (d)
- (a) $>$ (c) $>$ (e) $>$ (d) $>$ (b)
- (c) $>$ (d) $>$ (a) $>$ (e) $>$ (b)
- (a) $>$ (e) $>$ (b) $>$ (c) $>$ (d)

5. Dihedral angle of least stable conformer of ethane is: (2021)

- 180°
- 60°
- 0°
- 120°

6. Which of the following alkane cannot be made in good yield by Wurtz reaction? (2020)

- 2, 3-Dimethylbutane
- $n\text{-Heptane}$
- $n\text{-Butane}$
- $n\text{-Hexane}$

7. Which of the following is a free radical substitution reaction? (2020-Covid)

- Acetylene with HBr
- Methane with $\text{Br}_2/h\nu$
- Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{COO})_2$
- Benzene with $\text{Br}_2/\text{AlCl}_3$

8. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (2018)

- $\text{CH} \equiv \text{CH}$
- $\text{CH}_2 = \text{CH}_2$
- CH_4
- $\text{CH}_3 - \text{CH}_3$

9. With respect to the conformers of ethane, which of the following statements is true? (2017-Delhi)

- Both bond angles and bond length remains same
- Bond angle remains same but bond length changes
- Bond angle changes but bond length remains same
- Both bond angle and bond length change

10. The correct statement regarding the comparison of staggered and eclipsed conformations of ethane, is: (2016 - I)

- The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.
- The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain.
- The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain.
- The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain.

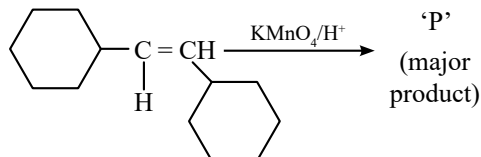
11. The number of structural isomers possible from the molecular formula C_3H_9N is: (2015 Re)
 a. 3 b. 4 c. 5 d. 2

Alkenes

12. Baeyer's reagent is : (2024 Re)

- a. Acidic potassium permanganate solution
 b. Acidic potassium dichromate solution
 c. Cold, dilute, aqueous solution of potassium permanganate
 d. Hot, concentrated solution of potassium permanganate

13. For the given reaction (2024)



'P' is

- a.
 b.
 c.
 d.

14. Which amongst the following compounds will show geometrical isomerism? (2023-Manipur)

- a. Pent-1-ene b. 2, 3-Dimethylbut-2-ene
 c. 2-Methylprop-1-ene d. 3, 4-Dimethylhex-3-ene

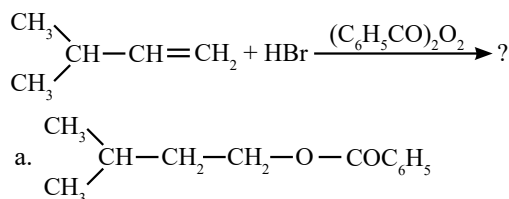
15. The **incorrect** method for the synthesis of alkenes is: (2022 Re)

- a. treating vicinal dihalides with Zn metal
 b. treatment of alkynes with Na in liquid NH_3
 c. heating alkyl halides with alcoholic KOH
 d. treating alkyl halides in aqueous KOH solution

16. Compound X on reaction with O_3 followed by Zn/H_2O gives formaldehyde and 2-methyl propanal as products. The compound X is: (2022)

- a. Pent-2-ene b. 3-Methylbut-1-ene
 c. 2-Methylbut-1-ene d. 2-Methylbut-2-ene

17. The major product of the following chemical reaction is: (2021)



- a.

- b.
 c.
 d.

18. An alkene on ozonolysis gives methanal as one of the product. Its structure is: (2020)

- a.
 b.
 c.
 d.

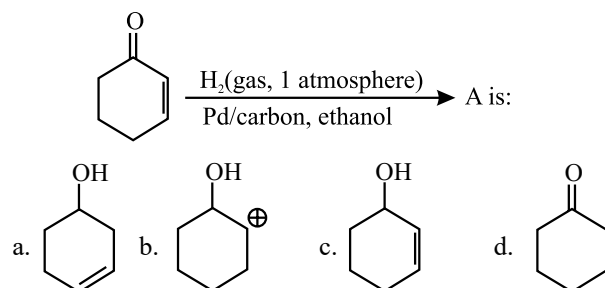
19. An alkene "A" on reaction with O_3 and $Zn - H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is: (2019)

- a.
 b.
 c.
 d.

20. The compound that will react most readily with gaseous bromine has the formula: (2016 - II)

- a. C_4H_{10} b. C_2H_4 c. C_3H_6 d. C_2H_2

21. The correct structure of the product A formed in the reaction: (2016 - II)

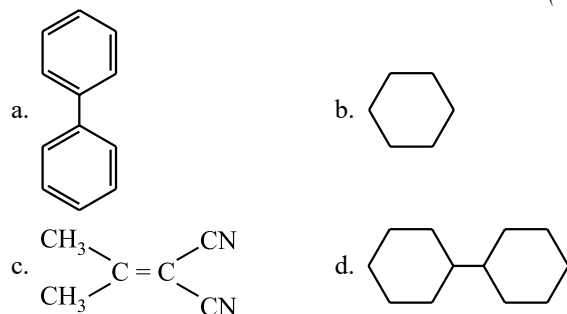


- a.
 b.
 c.
 d.

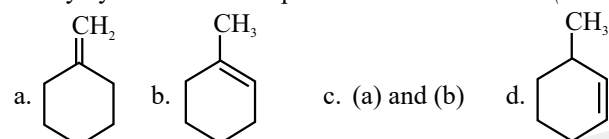
22. Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only elimination reaction? (2016 - II)

- a.
 b.
 c.
 d.

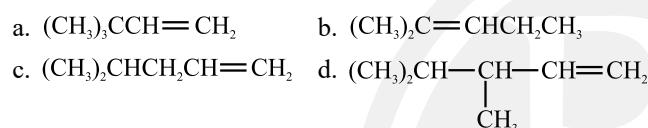
23. In which of the following molecules, all atoms are coplanar? (2016 - II)



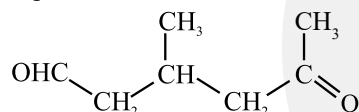
24. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is: (2015 Re)



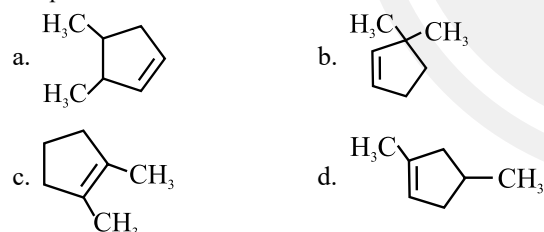
25. 2, 3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid? (2015 Re)



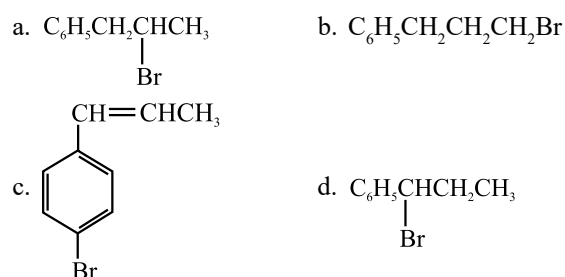
26. A single compound of the structure



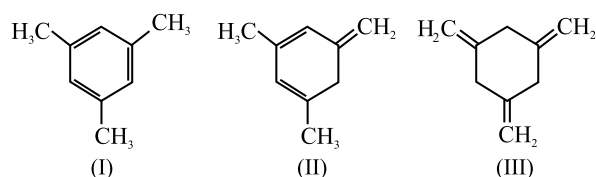
is obtainable from ozonolysis of which of the following cyclic compounds? (2015)



27. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3$ with HBr produces: (2015)



28. Given:



The enthalpy of hydrogenation of these compounds will be in the order as: (2015)

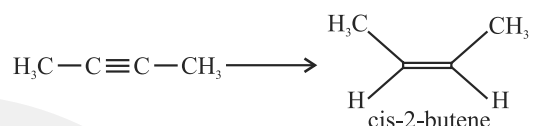
- a. $\text{III} > \text{II} > \text{I}$ b. $\text{II} > \text{III} > \text{I}$
- c. $\text{II} > \text{I} > \text{III}$ d. $\text{I} > \text{II} > \text{III}$

Alkynes

29. The number of sigma (s) and pi (p) bonds in pent-2-en-4-yne is (2019)

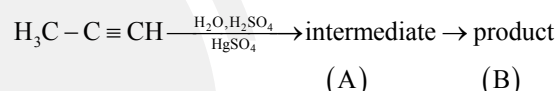
- a. 10 s bonds and 3p bonds
- b. 8 s bonds and 5p bonds
- c. 11 s bonds and 2p bonds
- d. 13 s bonds and no p bonds

30. The most suitable reagent for the following conversion, is: (2019)



- a. Na/liquid NH_3 b. H_2 , Pd/C, quinoline
- c. Zn/HCl d. $\text{Hg}^{2+}/\text{H}^+$, H_2O

31. Predict the correct intermediate and product in the following reaction (2017-Delhi)

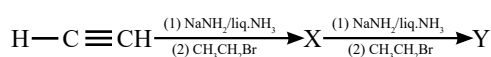


- a. A: $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$
- b. A: $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$
- c. A: $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$
- d. A: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$ B: $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$

32. Which one is the correct order of acidity? (2017-Delhi)

- a. $\text{CH}_3-\text{CH}_3 > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
- b. $\text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
- c. $\text{CH}\equiv\text{CH} > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}_3$
- d. $\text{CH}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_3-\text{CH}_3$

33. In the reaction: (2016 - I)



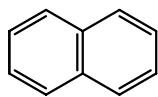
- a. X = 1-Butyne ; Y = 2-Hexyne
- b. X = 1-Butyne ; Y = 3-Hexyne
- c. X = 2-Butyne ; Y = 3-Hexyne
- d. X = 2-Butyne ; Y = 2-Hexyne

34. The pair of electron in the given carbanion, $\text{CH}_3\text{C}\equiv\text{C}^\ominus$, is present in which of the following orbitals? (2016 - I)

- a. sp b. d^2sp c. sp^3 d. sp^2

Aromatic Hydrocarbons

35. Consider the following compounds/species : (2023)



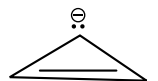
(I)



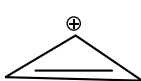
(II)



(III)



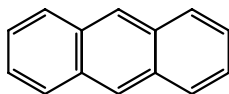
(IV)



(V)



(VI)

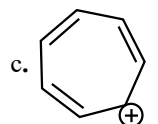
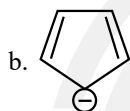
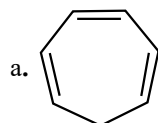


(VII)

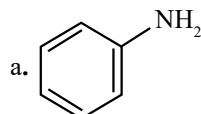
The number of compounds/species which obey Huckel's rule is

- a. 5 b. 4 c. 6 d. 2

36. Which compound amongst the following is **not an aromatic** compound? (2022)

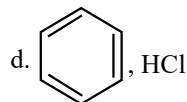


37. Which of the following is **suitable** to synthesize chlorobenzene? (2022)

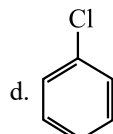
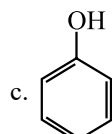
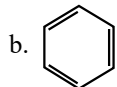
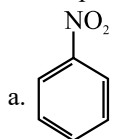


b. Benzene, Cl_2 , anhydrous FeCl_3

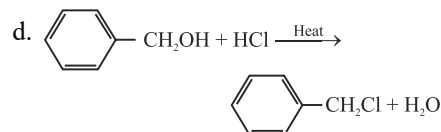
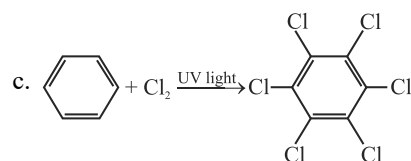
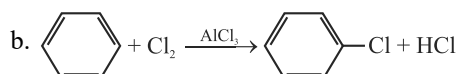
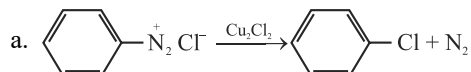
c. Phenol, NaNO_2 , HCl, CuCl



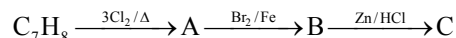
38. Which of the following compound is most reactive in electrophilic aromatic substitution? (2020-Covid)



39. Among the following, the reaction that proceeds through an electrophilic substitution, is: (2019)



40. The compound C_7H_8 undergoes the following reactions: (2018)



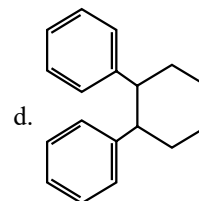
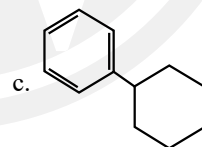
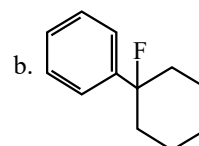
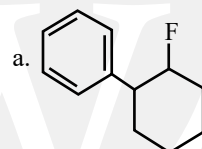
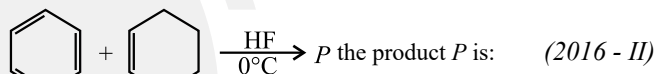
The product 'C' is

- a. m-bromotoluene b. o-bromotoluene
c. p-bromotoluene d. 3-bromo-2,4,6-trichlorotoluene

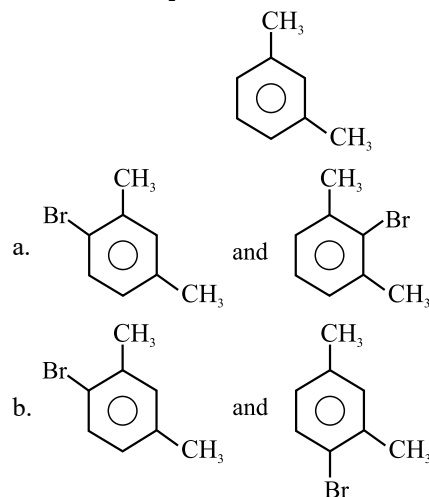
41. Which of the following can be used as the halide component for Friedel-Crafts reaction? (2016 - II)

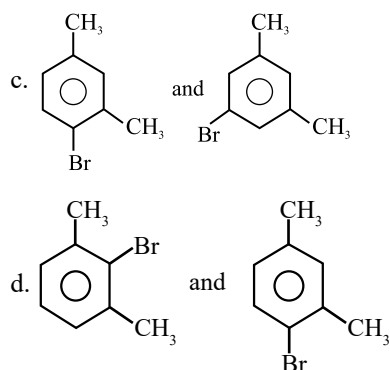
- a. Chloroethene b. Isopropyl chloride
c. Chlorobenzene d. Bromobenzene

42. In the given reaction,



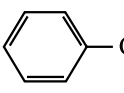
43. What products are formed when the following compound is treated with Br_2 in the presence of FeBr_3 ? (2014)





44. Which of the following compounds will not undergo Friedel-Craft reaction easily? (2013)

- a. Cumene b. Xylene
c. Nitrobenzene d. Toluene

45. The radical,  is aromatic because it has: (2013)

- a. 6 p-orbitals and 6 unpaired electrons
b. 7 p-orbitals and 6 unpaired electrons
c. 7 p-orbitals and 7 unpaired electrons
d. 6 p-orbitals and 7 unpaired electrons

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (c) | 4. (a) | 5. (c) | 6. (b) | 7. (b) | 8. (c) | 9. (a) | 10. (a) |
| 11. (b) | 12. (c) | 13. (d) | 14. (d) | 15. (d) | 16. (b) | 17. (d) | 18. (b) | 19. (c) | 20. (c) |
| 21. (d) | 22. (a) | 23. (a) | 24. (c) | 25. (a) | 26. (d) | 27. (d) | 28. (a) | 29. (a) | 30. (b) |
| 31. (b) | 32. (c) | 33. (b) | 34. (a) | 35. (b) | 36. (a) | 37. (b) | 38. (c) | 39. (b) | 40. (a) |
| 41. (b) | 42. (c) | 43. (b) | 44. (c) | 45. (a) | | | | | |